1.43 Trigonometry: Modelling in Degrees

Worked Solutions

1a)

$360° \div 2 = 180°$.

1b)

$360° \div 6 = 60°$.

1c)

$180° \div 1 = 180°$.

1d)

$360° \div \frac{1}{2} = 720°$.

1e)

$360° \div \frac{1}{4} = 1440°$.

1f)

$180° \div 2 = 90°$.

1g)

$360° \div \frac{360}{7} = 7°$.

1h)

$360° \div \frac{90}{13} = 52°$.

1i)

$360° \div \frac{72}{73} = 365°$.

1j)

$360° \div 15 = 24°$.

2a)

$y = \sin\left(\frac{360}{180}x\right) = \sin(2x)$.

2b)

$y = \sin\left(\frac{360}{1}x\right) = \sin(360x)$.

2c)

$y = \sin\left(\frac{360}{7}x\right)$.

2d)

$y = \sin\left(\frac{360}{52}x\right) = \sin\left(\frac{90}{13}x\right)$.

3a)

$y = 3\sin\left(\frac{360}{7}x\right)$.

3b)

$y = 2\sin\left(\frac{360}{1}x\right) = 2\sin(360x)$.

3c)

$y = 10\sin\left(\frac{360}{20}x\right) = 10\sin(18x)$.

3d)

$y = -5\sin\left(\frac{360}{30}x\right) = -5\sin(12x)$.

4a)

1.

4b)

1.

4c)

18.

4d)

5.

5a)

Max is 1, min is -1.

5b)

Max is $4 + 1 = 5$, min is $4 - 1 = 3$.

5c)

Max is $25 + 15 = 40$, min is $25 - 15 = 10$.

5d)

Max is $3 + 8 = 11$, min is $3 - 8 = -5$.

6a)

Period: 10.

$$360 \div 10 = 36$$

Amplitude: $\dfrac{5-1}{2} = 2.$

Translate $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.

$$y = 3 + 2\sin(36x).$$

6b)

Period: 30.

$$360 \div 30 = 12$$

Amplitude: $\dfrac{2.5-1.5}{2} = 0.5.$

Translate $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

$$y = 2 + 0.5\sin(12x).$$

6c)

Period: 20.

$$360 \div 20 = 18$$

Amplitude: $\dfrac{2--4}{2} = 3.$

Translate $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$.

$$y = -1 + 3\sin(18x).$$

6d)

Period: 7.

$$360 \div 7$$

Amplitude: $\dfrac{25-15}{2} = 5$ but -5 due to reflection.

Translate $\begin{pmatrix} 0 \\ 20 \end{pmatrix}$.

$$y = 20 - 5\sin\left(\frac{360}{7}x\right).$$

7a)

Max 8, Min 4.5.

Amplitude: $\dfrac{8-4.5}{2} = 1.75.$

$$8 - 1.75 = 6.25$$

$$y = 6.25 + 1.75\cos(x).$$

7b)

Change the period to match the yearly cycle:

$$y = 6.25 + 1.75\cos\left(\frac{360}{365}x\right).$$

8a)

Period: 1.

Max 120, Min 80.

Amplitude: $\dfrac{120-80}{2} = 20.$

$$120 - 20 = 100$$

$$y = 100 + 20\sin(360t).$$

8b)

Blood pressure will change depending on how much a person is moving.

9a)

$$20 + 6\sin(3 \times 5 - 10) = 20.52293446\ldots$$

so 21 mice.

9b)

$$20 + 6 = 26 \text{ mice}$$

$$20 - 6 = 14 \text{ mice}$$

9c)

$$3t - 10 = 90 \text{ so } t = \frac{100}{3}$$

Maximum occurs between day 33 and day 34.

$$3t - 10 = 270 \text{ so } t = \frac{280}{3}$$

Minimum occurs between day 93 and day 94.

10a)

$55 - 50 = 5$ and $55 + 50 = 105$.

So difference is £100.

10b)

$$360 \div \frac{72}{73} = 365 \text{ days}$$

So one year / 365 days later / 501 days after 1st January.

11a)

$5 - 1.7 = 3.3$ metres.

11b)

$$5 + 1.7\sin(25t - 330)^\circ = 6$$

$$\sin(25t - 330)^\circ = \frac{1}{1.7}$$

Find the first value:

$$25t - 330 = \arcsin\left(\frac{1}{1.7}\right) = 36.03187\ldots$$

Find the second value:

$$180 - 36.03187\ldots = 143.9681209\ldots$$

$$25t - 330 = 144.0$$

So

$$25t - 330 = 36.03187\ldots \Rightarrow t = 14.641275\ldots$$

$$25t - 330 = 143.9681209\ldots \Rightarrow t = 18.9587\ldots$$

Difference is 4.3 hours.

11c)

$$18.9587\ldots - \frac{360}{25} = 4.55872\ldots$$

So 4:34 am.

11d)

amplitude is $\frac{3.3}{2} = 1.65$ metres.

$$h = a + 1.65\sin(25t - 330)^\circ$$

High tide for Laiyla's model: $5 + 1.7 = 6.7$ metres.

$$6.7 - 2.55 = 4.15 \text{ metres}$$

$$a + 1.65 = 4.15 \ \ \therefore a = 2.5$$

$$\therefore h = 2.5 + 1.65\sin(25t - 330)^\circ.$$